Transfer: momentum and impulse

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Learning target: Apply momentum reasoning to new collisions and deflection cases.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 11.*
Shift becomes 3 times as large: shift=Δv×time. The simplified model omits orbital curvature/gravity and uncertainties.

02. Worked example: explain the model
Arrow goes from +6 to 0 kg·m/s; rectangle area -6 N·s.

03. Read and interpret the model
Total momentum 4 kg·m/s; after both move 2 m/s right. Kinetic energy is not conserved.

04. Apply the physics
Momentum-change magnitude = 12 kg·m/s. Average stopping-force magnitude = 12/5 = 2.4 N, directed left.

05. Explain your reasoning
Compare total kinetic energy; must also match for elastic.

06. Review: connect an earlier idea *Review: Unit 3, Day 2.*
24 kg·m/s left.

07. Exit check
20 m; constant change/straight line approximation ignores orbital physics.

08. STRETCH: test the model
Examples 0.25 s at -24 N and 1 s at -6 N; same -6 N·s.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 11.*
$\Delta v \approx 0.0006$ m/s; impulse$\approx 4.2\times10^{5}$ N·s. Uses constant added transverse velocity over a straight-line comparison.

02. Worked example: explain the model
Arrow goes from +6 to 0 kg·m/s; rectangle area -6 N·s.

03. Read and interpret the model
Total momentum 4 kg·m/s; after both move 2 m/s right. Kinetic energy is not conserved.

04. Apply the physics
vf 3 m/s; initial KE 40.5 J; final 13.5 J; 27 J transformed.

05. Explain your reasoning
Compare total kinetic energy; must also match for elastic.

06. Review: connect an earlier idea *Review: Unit 3, Day 2.*
v=-6 m/s; after reversal p=+24 kg·m/s.

07. Exit check
Impulse 6 N·s positive; initial p -2, final p 4; vf +2 m/s.

08. STRETCH: test the model
Impulse -3.2 N·s; average -40 N; triangle peak magnitude 80 N.